

Skills Summary

Coordinate Plane, Line Plots, and Numerical Patterns

Use this reference while completing your worksheet or when studying.

Skill 1 — Two patterns make ordered pairs

Rule: run both patterns from the same starting point, then pair *matching terms* — 1st with 1st, 2nd with 2nd. Write each pair as (x, y) and plot it: across first, then up.

When both patterns start at zero, the y -value is always the same multiple of the x -value, and that multiplier is

$$\text{multiplier} = \frac{\text{step size of the } y \text{ pattern}}{\text{step size of the } x \text{ pattern}}$$

Example: Pattern M adds 3 each time, Pattern N adds 6 each time, both from zero. Find the N -value paired with the M -value 8.

Step 1. Both patterns start together, so matching terms grow in a fixed ratio — find the multiplier from the two step sizes instead of writing out the lists.

Step 2. Multiplier = $6 \div 3 = 2$, so $N = 2 \times 8 = 16$, and the ordered pair is $(8, 16)$.
 $\Rightarrow N = 16$

Try it: Pattern M adds 2 each time, Pattern N adds 10 each time, both from zero. Find the N -value paired with the M -value 7.

check: 58

Skill 2 — Summarising a line plot

Read the plot twice. The number *under* a stack is the value; the number of $\times$ marks *in* the stack is how many times that value happened.

- **Total** = add (value $\times$ stack height) for every stack.
- **Mean** = total $\div$ number of marks (not number of stacks).
- **Median** = middle value, counting *marks* from the left.

Example: A line plot shows six values: 1, 2, 3, 3, 4, 5. Find the mean.

Step 1. A mean needs a total and a count, and the plot gives both — add the values, then divide by how many marks there are.

Step 2. $1 + 2 + 3 + 3 + 4 + 5 = 18$, and there are six marks, so the mean is $18 \div 6 = 3$.
 $\Rightarrow \text{mean} = 3$

Try it: A line plot shows six values: 2, 2, 4, 4, 5, 7. Find the mean.

check: 4

Skill 3 — Running a rule on data you read

Two-piece problems: one piece is a number you must *read or compute* from the plot or table; the other is a *rule* stated in words.

Write the read value down on its own line first, then apply the rule to it. “Points for every book” multiplies; “beyond” or “more than” subtracts first; “each value goes up by” shifts the mean by that same amount.

Example: A class read 30 books in all. The club gives 4 points for every book. How many points did the class earn?

Step 1. The plot supplies the book total and the rule supplies the points per book — write the read value first, then apply the rule to it.

Step 2. Read value: 30 books. Rule: 4 points a book, so $4 \times 30 = 120$. $\Rightarrow$ **120** points

Try it: A class read 25 books in all, and the club gives 3 points for every book. How many points did the class earn?

check: 75

(!) Watch out: on a line plot the $\times$ marks count *how many people*, never *how much*. Counting marks answers “how many students”; adding value times stack height answers “how many books”. And when pairing two patterns, always pair terms in the *same position* — one step out of step is the most common wrong ordered pair there is.